

Long Lived Particle Searches With the ATLAS Detector Using Ionization Energy
Loss and Disappearing Tracks.

by

Nathan Dale Young

A dissertation accepted and approved in partial fulfillment of the
requirements for the degree of
Doctor of Philosophy
in Physics

Dissertation Committee:

Stephanie Majewski, Chair

Graham Kribs, Core Member

Laura Jeanty, Advisor

Ben Elias, Institutional Representative

University of Oregon

May 2026

© 2026 Nathan Dale Young
All rights reserved.

DISSERTATION ABSTRACT

Nathan Dale Young

Doctor of Philosophy in Physics

Title: Long Lived Particle Searches With the ATLAS Detector Using Ionization Energy Loss and Disappearing Tracks.

CURRICULUM VITAE

NAME OF AUTHOR: Nathan Dale Young

GRADUATE AND UNDERGRADUATE SCHOOLS ATTENDED:

University of Oregon, Eugene, OR, USA

University of Arizona, AZ, USA

Pima Community College, AZ, USA

DEGREES AWARDED:

Doctor of Philosophy, Physics, 2026, University of Oregon

Master of Science, Physics, 2022, University of Oregon

Bachelor of Science, Physics, 2022, University of Arizona

Associate of Science, Physics, 2018, Pima Community College

AREAS OF SPECIAL INTEREST:

PROFESSIONAL EXPERIENCE:

GRANTS, AWARDS AND HONORS:

PUBLICATIONS:

ACKNOWLEDGEMENTS

Here is an acknowledgment

To so-and-so...

TABLE OF CONTENTS

Chapter	Page
I. INTRODUCTION	11
II. OVERVIEW AND MOTIVATION	12
2.1. Background Motivation	12
2.2. Supersymmetry	12
2.2.1. Coannihilation Stau	12
2.2.2. GMSB Stau	12
III. EXPERIMENTAL SETUP	13
3.1. The Large Hadron Collider	13
3.2. The ATLAS Detector	14
3.2.1. Inner Detector	16
3.2.1.1. Pixel Detector	17
3.2.1.2. Semiconductor Tracker	19
3.2.1.3. Transition Radiation Tracker	19
3.2.2. Calorimeters	20
3.2.2.1. Electromagnetic Calorimeter	20
3.2.2.2. Hadronic Calorimeter	21
3.2.3. Muon Spectrometer	22
IV. EXPERIMENTAL UPGRADES	24
4.1. Phase 1 upgrades	25
4.2. Phase 2 upgrades	25
4.2.1. ATLAS ITk	25
4.2.1.1. Pixel Detector	26

Chapter	Page
V. OBJECT RECONSTRUCTION	29
VI. DISAPEARING TRACKS	30
VII. DISAPEARING TRACKS WITH HIGH IONIZATION LOSS	31
VIII.CONCLUSION	32
APPENDICES	
A. THE FIRST APPENDIX	33
A.1. Appendix One Section One	33
A.1.1. Chapter four section one sub-section one	33
B. THE SECOND APPENDIX	34
B.1. Appendix Two Section One	34
B.1.1. Chapter two section one sub-section one	34
REFERENCES CITED	35

LIST OF FIGURES

Figure		Page
1.	Schematic of the LHC accelerator complex[5], showing the sequence of accelerators used to prepare and inject proton beams into the LHC ring. Image credit: CERN	14
2.	Schematic view of the ATLAS detector[2], showing the main subsystems arranged around the interaction point. Image credit: CERN	15
3.	Schematic view of the Inner Detector[2], showing the beam pipe, pixel barrel, SCT barrel, and TRT. Image credit: CERN	17
4.	Schematic view of silicion microstrip detector. credit:	20
5.	Schematic view of the Calorimeters[2], showing the electromagnetic and hadronic calorimeters. Image credit: CERN	21
6.	Schematic layout of the ATLAS Muon Spectrometer showing the barrel and endcap regions along with the locations of the different detector technologies used for muon tracking and triggering.	23
7.	View of ITk layers and locations in r-z plane.	28

LIST OF TABLES

Table	Page
1. Stave locations	27
2. Ring locations	27

CHAPTER I
INTRODUCTION

CHAPTER II

OVERVIEW AND MOTIVATION

2.1 Background Motivation

2.2 Supersymmetry

2.2.1 Coannihilation Stau.

2.2.2 GMSB Stau.

CHAPTER III

EXPERIMENTAL SETUP

This chapter describes the experimental setup used to collect the data for the long-lived particle searches presented in Sections VI and VII. All results presented in this thesis are based on data collected by the ATLAS detector at the Large Hadron Collider (LHC) at CERN.

3.1 The Large Hadron Collider

The Large Hadron Collider (LHC) is currently the world's largest and highest-energy particle accelerator, with a total circumference of approximately 27 km. It is located about 100 m underground, straddling the border between Switzerland and France near Geneva. The LHC accelerates protons to nearly the speed of light using radio-frequency (RF) cavities and superconducting magnets, and brings counter-rotating proton beams into collision at four interaction points evenly spaced around the accelerator ring. At each interaction point resides one of the four main experiments: ATLAS, CMS, LHCb, and ALICE. The center-of-mass energy of proton–proton collisions is 13 TeV during Run 2 and 13.6 TeV during Run 3.

Protons are produced from hydrogen gas and accelerated through a sequence of injector machines prior to entering the LHC. Since hydrogen is initially in molecular form, electrons are first added to form H^- ions. These ions are then passed through a stripping foil, removing the electrons and producing protons (H^+). The protons are initially accelerated by RF cavities and focused using quadrupole magnets in the Linear Accelerator 4 (LINAC4), reaching an energy of 160 MeV.

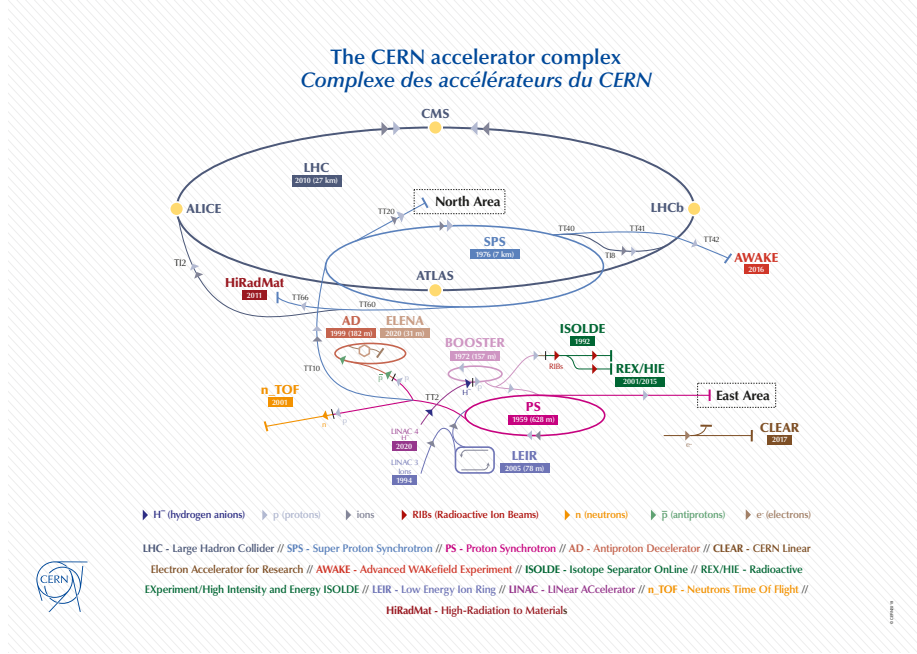


Figure 1. Schematic of the LHC accelerator complex[5], showing the sequence of accelerators used to prepare and inject proton beams into the LHC ring. Image credit: CERN .

Subsequently, the proton beam is injected into the Proton Synchrotron Booster (PSB), where it is further accelerated to 2 GeV. From there, the beam is transferred through the Proton Synchrotron (PS) and the Super Proton Synchrotron (SPS), where it undergoes additional acceleration and bunch shaping before final injection into the LHC. A detailed description of the injector chain performance can be found in Ref. [1].

Finally, the prepared proton bunches are injected into the LHC, where they are accelerated to their final energies and brought into collision at the four interaction points.

3.2 The ATLAS Detector

The ATLAS detector is a general purpose particle detector located at one of the four interaction points of the Large Hadron Collider (LHC). It is approximately

46 m long, 25 m in diameter, weighs about 7000 tons, and consists of multiple subsystems arranged concentrically around the interaction point.

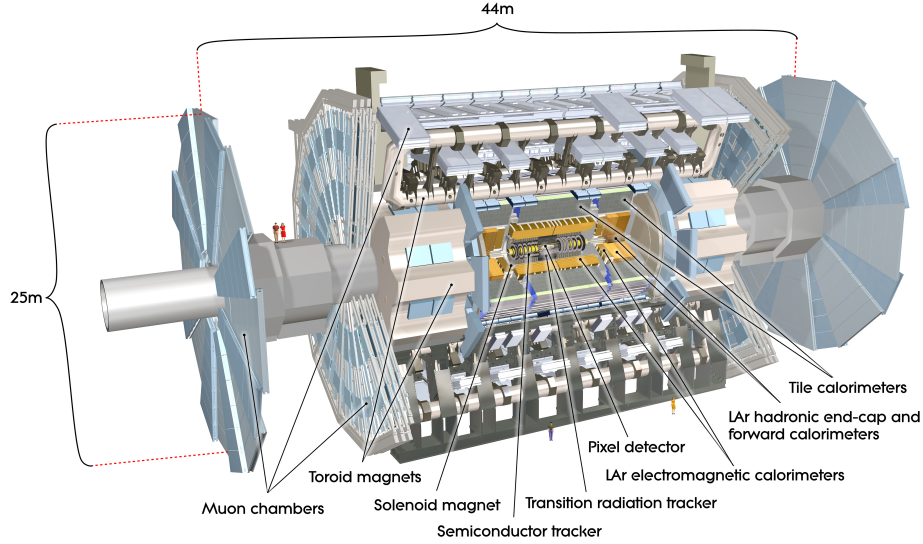


Figure 2. Schematic view of the ATLAS detector[2], showing the main subsystems arranged around the interaction point. Image credit: CERN .

Starting from the interaction point and moving radially outward, the main subsystems are the Inner Detector (ID), the central solenoid magnet, the electromagnetic and hadronic calorimeters, the toroidal magnet system, and the Muon Spectrometer (MS). Each subsystem is designed to measure specific properties of particles produced in proton–proton collisions.

The Inner Detector is the closest subsystem to the interaction point and is responsible for reconstructing the trajectories of charged particles. It provides precise measurements of track positions and momenta within a 2 T magnetic field generated by the central solenoid, which surrounds the ID. Outside the solenoid lies the electromagnetic calorimeter, which measures the energies of electrons and photons through the development of electromagnetic showers. Surrounding this is the hadronic calorimeter, designed to measure the energies of hadrons via hadronic

showering, working in conjunction with the electromagnetic calorimeter to provide a complete measurement of the event energy flow.

Enclosing the calorimeters is the toroidal magnet system, which supplies the magnetic field required for the Muon Spectrometer. The MS forms the outermost layer of ATLAS and is optimized for the detection and momentum measurement of muons. Due to their relatively large mass and weak interactions with matter, muons typically traverse the inner detector and calorimeters with minimal energy loss, making the MS essential for their identification. The MS employs a combination of tracking chambers and magnetic fields to measure the curvature of muon trajectories, enabling precise momentum determination.

The ATLAS detector uses a right handed coordinate system. The origin is at the nominal interaction point with the Z axis along the beam pipe. The Y axis points upwards and the X axis points to the center of the LHC ring. In ATLAS the normal angles θ and ϕ are used where θ is the polar angle measured from the positive z axis and ϕ is the azimuthal angle measured in the x-y plane from the x axis. One important variable used in ATLAS is pseudorapidity, η , which is defined as $\eta = -\ln(\tan(\theta/2))$ where θ is the polar angle measured from the positive z axis. Pseudorapidity is useful for ATLAS as it is invariant under Lorentz boosts along the beam axis with the assumption that the particles momentum is much larger than mass, $pc \gg mc^2$.

3.2.1 Inner Detector. The inner detector (ID) is a marvel of technology and innovation that was used in collaboration with other detector systems on ATLAS to not only discover a new particle but to confirm that it was a scalar Higgs Boson. The ID is at the center of the ATLAS detector and measures momentum, position, impact parameter, charge, and some particle identification

including via ionization loss, of charged particles that pass through it. The ID consists of three main components: the Pixel Detector, the Semiconductor Tracker (SCT), and the Transition Radiation Tracker (TRT).

Particles that transverse the Pixel Detector, the SCT, and the TRT have momentum resolution which can be parameterized as

$$\frac{\sigma_{p_T}}{p_T} = 0.05\% \times p_T [\text{GeV}] \oplus 1\%. \quad (3.1)$$

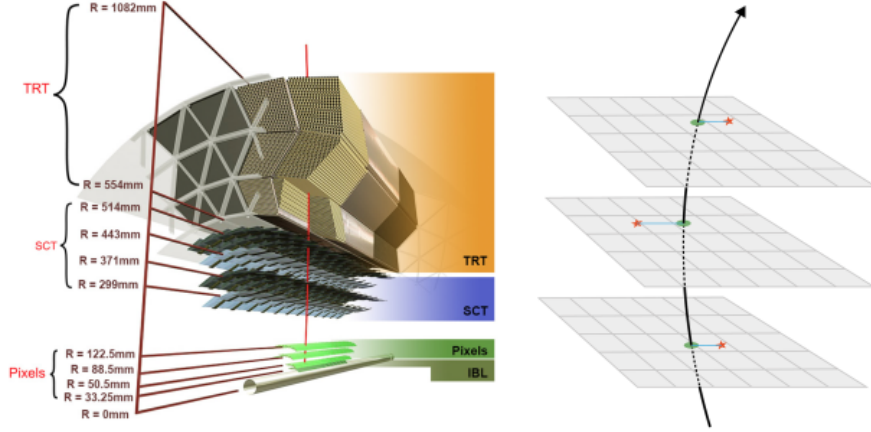


Figure 3. Schematic view of the Inner Detector[2], showing the beam pipe, pixel barrel, SCT barrel, and TRT. Image credit: CERN .

3.2.1.1 Pixel Detector. The Pixel Detector is located closest to the interaction point and is composed of multiple layers of silicon pixel sensors, providing charged-particle tracking over radial distances from 33.5 mm to 122.5 mm. It consists of the Insertable B-Layer (IBL), three barrel layers, and three endcap disks on each side of the detector. This configuration provides up to four space-point measurements for charged particles within a pseudorapidity coverage of $|\eta| < 2.5$.

The innermost pixel layer, known as the Insertable B-Layer (IBL), was installed during the first long shutdown of the LHC to enhance tracking and vertexing performance. The IBL is positioned at a radius of 33.5 mm from the beamline, enabling high-precision measurements close to the interaction point. It is constructed from 14 staves, each containing 32 front-end chips mounted on silicon sensors.

Two sensor technologies are employed in the IBL: planar sensors and 3D sensors. Planar sensors are used in the central region of each stave, while 3D sensors are utilized in the forward regions due to their superior radiation tolerance. Each sensor is segmented into pixels with dimensions of $50\text{ }\mu\text{m}$ in ϕ and $250\text{ }\mu\text{m}$ in z . Each stave contains 1280 columns and 336 rows of pixels, resulting in approximately 12.8 million pixels in the full IBL.

The remaining three barrel layers of the Pixel Detector, referred to as the B-layer, Layer 1, and Layer 2, are located at radii of 50.5 mm, 88.5 mm, and 122.5 mm, respectively. In these layers, planar silicon sensors with a thickness of $250\text{ }\mu\text{m}$ are used, with pixel dimensions of $50\text{ }\mu\text{m}$ in ϕ and $300\text{ }\mu\text{m}$ in z .

The pixel detector is capped by 3 endcaps on either side of the collision point at z distances of 495, 580, 650

A pixel module consists of a silicon sensor bump-bonded to dedicated readout chips. The sensors are based on reverse-biased silicon diodes. Applying a reverse bias voltage creates a depleted region within the sensor volume. Charged particles traversing this depletion region generate approximately 90 electron-hole pairs per micrometer, with each pair requiring about 3.16 eV to be produced. The resulting charge carriers drift under the applied electric field, inducing a measurable signal on the electrodes.

Direct measurement of the collected charge would require power-intensive analog-to-digital conversion. Instead, the pixel readout employs a digital scheme in which the induced signal is first amplified and shaped, and then compared to a predefined threshold. The duration for which the signal remains above this threshold, referred to as the Time-over-Threshold (ToT), is proportional to the deposited charge. Due to the non-linear response of the electronics, calibration is performed by injecting known charges and measuring the corresponding ToT response.

3.2.1.2 *Semiconductor Tracker.* The Semiconductor Tracker (SCT) is located outside the Pixel Detector and provides precision tracking at radial distances between approximately 299 mm and 560 mm from the beamline, covering a pseudorapidity range of $|\eta| < 2.5$. The SCT is a silicon microstrip detector designed to complement the Pixel Detector by extending the tracking lever arm and improving momentum resolution.

The SCT consists of 4088 silicon strip modules arranged in four barrel layers and two endcaps, each comprising nine disks. Each module contains two back-to-back silicon strip sensors mounted with a small stereo angle, enabling three-dimensional space-point reconstruction.

3.2.1.3 *Transition Radiation Tracker.* The Transition Radiation Tracker (TRT) forms the outermost component of the Inner Detector. It is composed of straw tubes filled with a xenon-based gas mixture, which serve both as tracking detectors and as transition radiation detectors for electron identification.

As charged particles traverse the straw tubes, they ionize the gas, and the resulting electrons drift toward the central anode wires where the signals are read out. Each straw has a diameter of 4 mm and can provide up to approximately 36

7 Track detectors

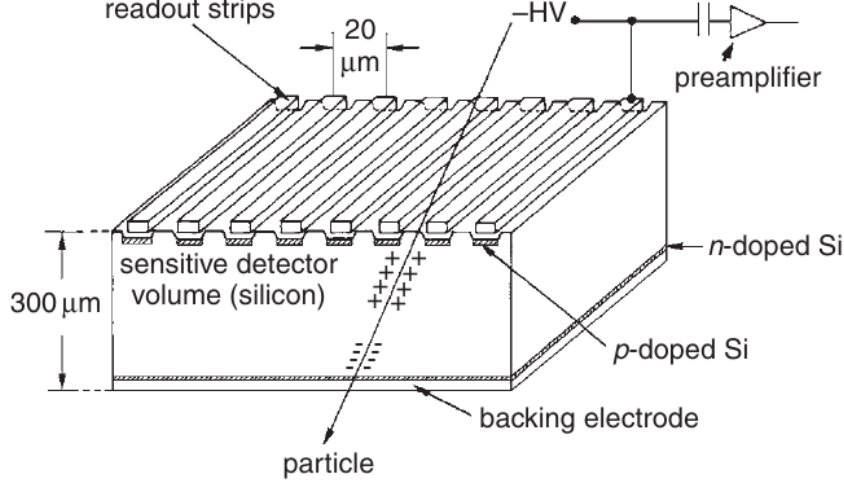


Figure 4. Schematic view of silicon microstrip detector. credit:

measurements per track, significantly increasing the number of tracking points in the outer region of the Inner Detector.

A limitation of the TRT is that it provides position measurements only in the R - ϕ plane. Nevertheless, when combined with the Pixel Detector and SCT, the TRT enhances pattern recognition and improves track reconstruction, particularly in dense environments.

3.2.2 Calorimeters. ATLAS employs two main calorimeter systems: the electromagnetic calorimeter and the hadronic calorimeter. Together, these systems are designed to absorb and measure the energies of particles emerging from the interaction point after they traverse the Inner Detector.

3.2.2.1 Electromagnetic Calorimeter. The electromagnetic calorimeter is a sampling calorimeter that uses lead as the absorber material and liquid argon as the active medium. Incident electrons and photons initiate

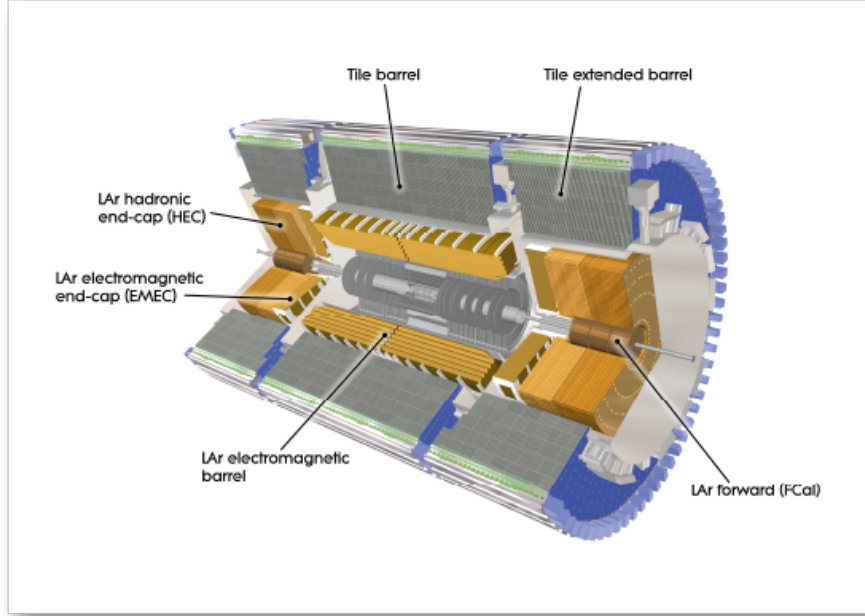


Figure 5. Schematic view of the Calorimeters[2], showing the electromagnetic and hadronic calorimeters. Image credit: CERN .

electromagnetic showers in the lead absorbers, with the resulting ionization signals collected in the liquid argon layers to provide precise energy measurements.

The electromagnetic calorimeter is divided into a barrel section and two endcap sections. The barrel region covers pseudorapidities of $|\eta| < 1.475$, while the endcap regions extend the coverage to $1.375 < |\eta| < 3.2$.

3.2.2.2 Hadronic Calorimeter. The hadronic calorimeter is also a sampling calorimeter, employing steel and copper as absorber materials and scintillating tiles and liquid argon as active media. It is designed to measure the energies of hadrons through the development of hadronic showers.

Similar to the electromagnetic calorimeter, the hadronic calorimeter is divided into a barrel section and two endcap sections. The barrel region covers $|\eta| < 1.4$, while the endcap regions extend the coverage to $1.4 < |\eta| < 4.8$. In the

forward and endcap regions, liquid argon technology is used to accommodate the higher radiation levels.

3.2.3 Muon Spectrometer. Due to their relatively large mass and weak interactions with matter, muons typically traverse the Inner Detector and calorimeters while depositing only small amounts of ionization energy. As a result, the outermost subsystem of ATLAS is dedicated to reconstructing the momentum and trajectory of these weakly interacting charged particles.

The Muon Spectrometer is constructed using five detector technologies: Thin Gap Chambers (TGCs), Resistive Plate Chambers (RPCs), Monitored Drift Tubes (MDTs), Small-Strip Thin Gap Chambers (sTGCs), and Micromegas. The majority of the precision tracking measurements are provided by the Monitored Drift Tubes.

The Muon Spectrometer consists of a barrel region and two endcap regions. The barrel region covers the pseudorapidity range $|\eta| < 1.2$, while the endcap regions extend the coverage to $1.0 < |\eta| < 2.7$. A schematic view of the Muon Spectrometer and its various detector components is shown in Figure 6.

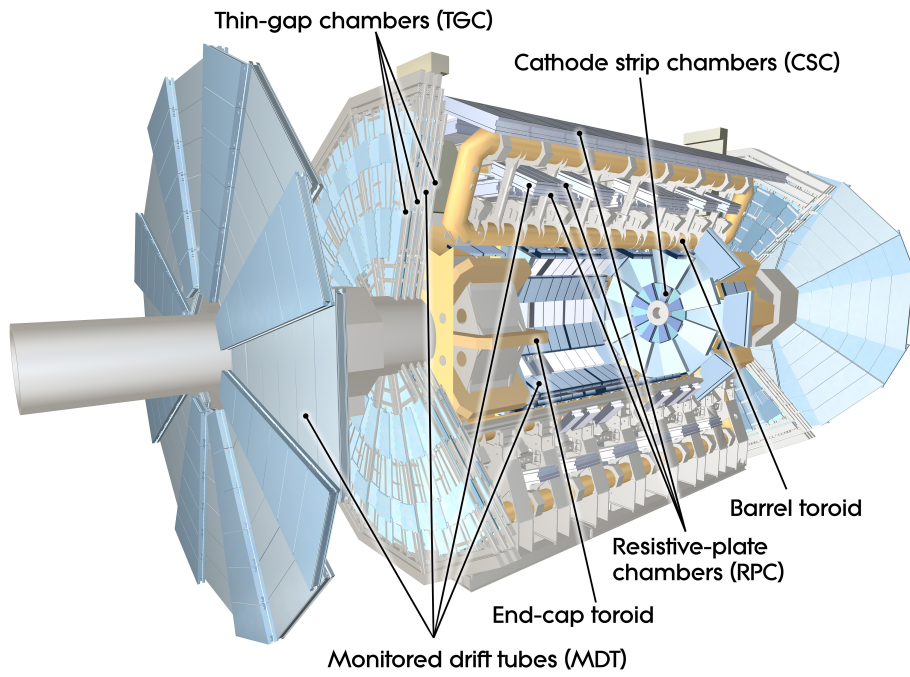


Figure 6. Schematic layout of the ATLAS Muon Spectrometer showing the barrel and endcap regions along with the locations of the different detector technologies used for muon tracking and triggering.

CHAPTER IV

EXPERIMENTAL UPGRADES

The ATLAS detector is currently operating beyond its original design specifications from the early 2000s. As the LHC has steadily increased its delivered luminosity, the ATLAS detector has required a series of upgrades to cope with higher data rates, increased detector occupancy, and elevated radiation levels.

Beginning in approximately 2030, the LHC will enter the High-Luminosity phase (HL-LHC). During this period, the instantaneous luminosity is expected to increase by up to an order of magnitude relative to the original LHC design, reaching values of

$$\mathcal{L} = 5 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}.$$

At these luminosities, an average of approximately 200 proton–proton interactions per bunch crossing is anticipated, compared to the typical 20–50 interactions observed during Run 2. This dramatic increase in pileup presents significant challenges for event reconstruction, triggering, and data acquisition.

To meet these challenges and continue exploring the particle physics phenomena at the HL-LHC, the ATLAS detector is undergoing extensive upgrades. These upgrades are designed to improve radiation tolerance, increase detector granularity, enhance readout bandwidth, and maintain or improve physics performance under extreme operating conditions. Major upgrade components include a complete replacement of the Inner Detector with a new all-silicon Inner Tracker (ITk), upgrades to the Muon Spectrometer, new front-end and readout electronics for the calorimeters, and substantial improvements to the trigger and data acquisition systems.

The ATLAS upgrade program is divided into two phases. Phase 1 upgrades were completed during the Long Shutdown 2 period and commissioned by 2024. Phase 2 upgrades are scheduled for completion by 2030, prior to the start of HL-LHC operations.

4.1 Phase 1 upgrades

The Phase 1 upgrade program focused on improving detector performance and trigger capabilities in preparation for Run 3 of the LHC. Key components of this phase included upgrades to the Muon Spectrometer, the introduction of new trigger electronics, and improvements to detector readout systems. These upgrades enabled ATLAS to operate efficiently at higher luminosities and increased event rates during Run 3.

4.2 Phase 2 upgrades

The Phase 2 upgrade program is designed to prepare the ATLAS detector for sustained operation during the HL-LHC era and is scheduled for completion in 2030. The most significant component of this upgrade is the complete replacement of the current Inner Detector with a new all-silicon Inner Tracker (ITk).

The ITk will provide substantially higher granularity, improved radiation hardness, and faster readout capabilities compared to the existing Inner Detector. It is designed to maintain excellent tracking and vertexing performance in the presence of extreme pileup and radiation levels expected at the HL-LHC. The design and performance of the ITk are described in detail in the ATLAS Technical Design Reports [4, 3], and the ITk forms the primary focus of this chapter.

4.2.1 ATLAS ITk. The primary goal of the ITk upgrade is to maintain or improve charged-particle tracking performance in the high-luminosity environment of the HL-LHC. Two major challenges must be addressed to achieve

this goal: the substantially increased detector occupancy due to high pileup, and the extreme radiation levels expected over the lifetime of the detector.

The innermost layers of the ITk are expected to experience radiation fluences exceeding $2 \times 10^{16} \text{ n}_{\text{eq}}/\text{cm}^2$ and total ionizing doses of up to approximately 1.7 Grad. To operate reliably under these conditions, the ITk employs radiation hard silicon sensor technologies, advanced readout electronics, and a detector layout optimized for performance and longevity. The first layer of the ITk is also designed to be replaceable, allowing for potential replacement during the HL-LHC era.

ITk is split into two main systems: the Pixel Detector and the Strip Detector.

4.2.1.1 *Pixel Detector.* The combined effects of integrated luminosity and the planned increase of instantaneous luminosity result in the ID no longer be sufficient for tracking and vertexing beyond run 3 in ATLAS and will be replaced. The ID is being replaced with a new all-silicon Inner Tracker (ITk). ITk is designed to accommodate the 200 collisions per bunch crossing expected in the high luminosity LHC while providing readout at 1MHz so that the tracking can be used in L0-L1 triggers. The core of the ITk is the pixel module which detects the passage of charged particles. The pixel module is made of a silicon detector bonded directly to the front-end electronics for readout. The pixel modules will be attached to local supports either rings or staves made of carbon fiber laminates and high thermal-conductive foam core with titanium pipes traversing the center for cooling. These rings and staves are mounted into shells that will provide outer structure for ITk as can be seen in figure 1.

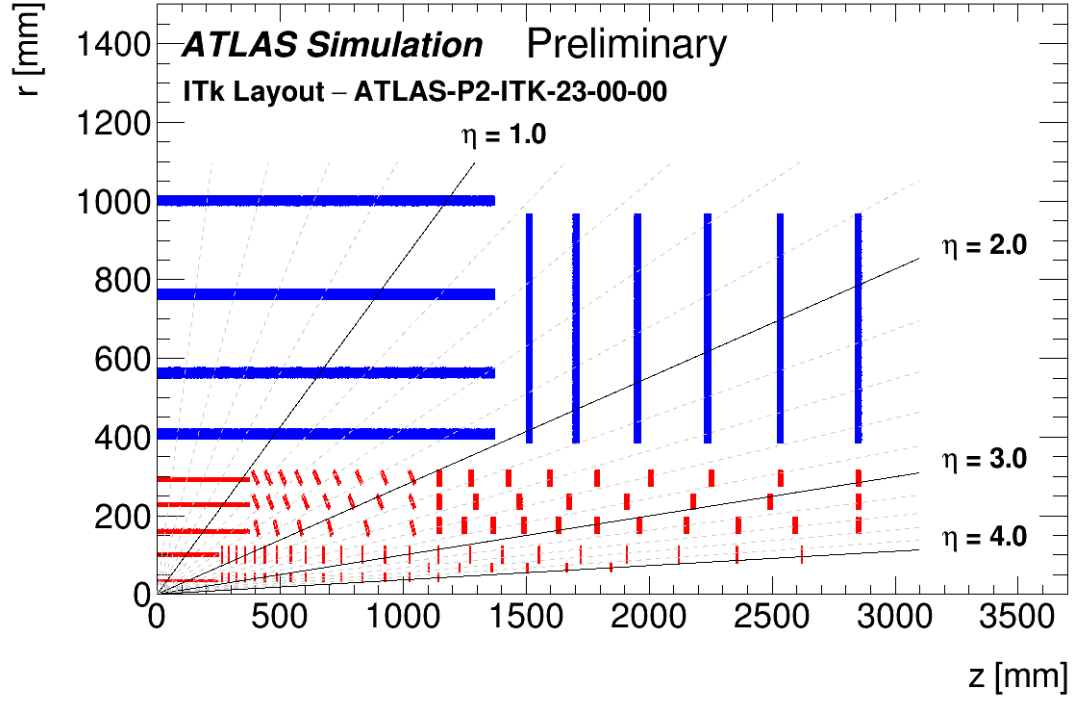
The US ATLAS program is responsible for the two inner most layers of ITk called the Inner System. The Inner System consists of 2348 modules. SLAC National Laboratory is responsible for the assembly of the inner detector.

Layer	Radius [mm]	Z Range [mm]	Pixel Size [μm]	Sensor Thickness
L0	34	± 245	25x100	270
L1	99	± 245	50x50	100
L2	160	± 372	50x50	150
L3	228	± 372	50x50	150
L4	291	± 372	50x50	150

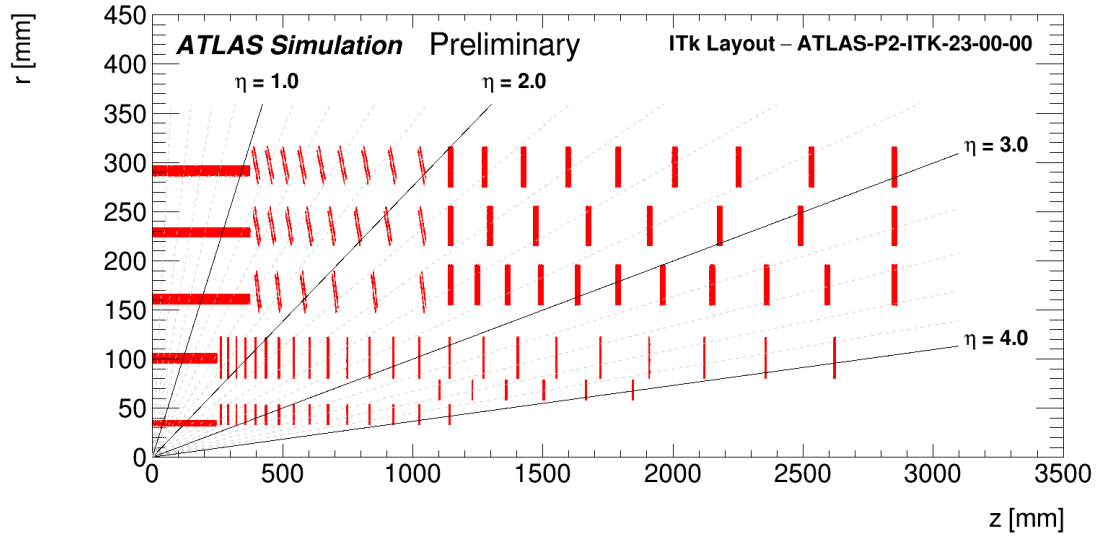
Table 1. Stave locations

Layer	Radius [mm]	Pixel Size [μm]	Sensor Thickness [μm]
0	33.20	25x100	270
.5	58.7	25x100	100
1	80.00	50x50	150
2	154.50	50x50	150
3	214.55	50x50	150
4	274.60	50x50	150

Table 2. Ring locations



(a) ITk layout



(b) Inner Detector layout

Figure 7. View of ITk layers and locations in r - z plane.

CHAPTER V

OBJECT RECONSTRUCTION

CHAPTER VI
DISAPPEARING TRACKS

CHAPTER VII

DISAPPEARING TRACKS WITH HIGH IONIZATION LOSS

CHAPTER VIII

CONCLUSION

APPENDIX A

THE FIRST APPENDIX

A.1 Appendix One Section One

A.1.1 Chapter four section one sub-section one.

APPENDIX B

THE SECOND APPENDIX

B.1 Appendix Two Section One

B.1.1 Chapter two section one sub-section one. What?

REFERENCES CITED

- [1] Hannes Bartosik and Giovanni Rumolo. Performance of the lhic injector chain after the upgrade and potential development, 2022.
- [2] ATLAS Collaboration. Computer generated image of the whole atlas detector. <https://cds.cern.ch/record/1095924>, 2008.
- [3] ATLAS Collaboration. Technical design report for the atlas inner tracker pixel detector. <https://cds.cern.ch/record/2285585>, 2017.
- [4] ATLAS Collaboration. Technical design report for the atlas inner tracker strip detector. <https://cds.cern.ch/record/2257755/>, 2017.
- [5] ATLAS Collaboration. The cern accelerator complex, complexe des accélérateurs du cern. <https://cds.cern.ch/record/2693837>, 2019.